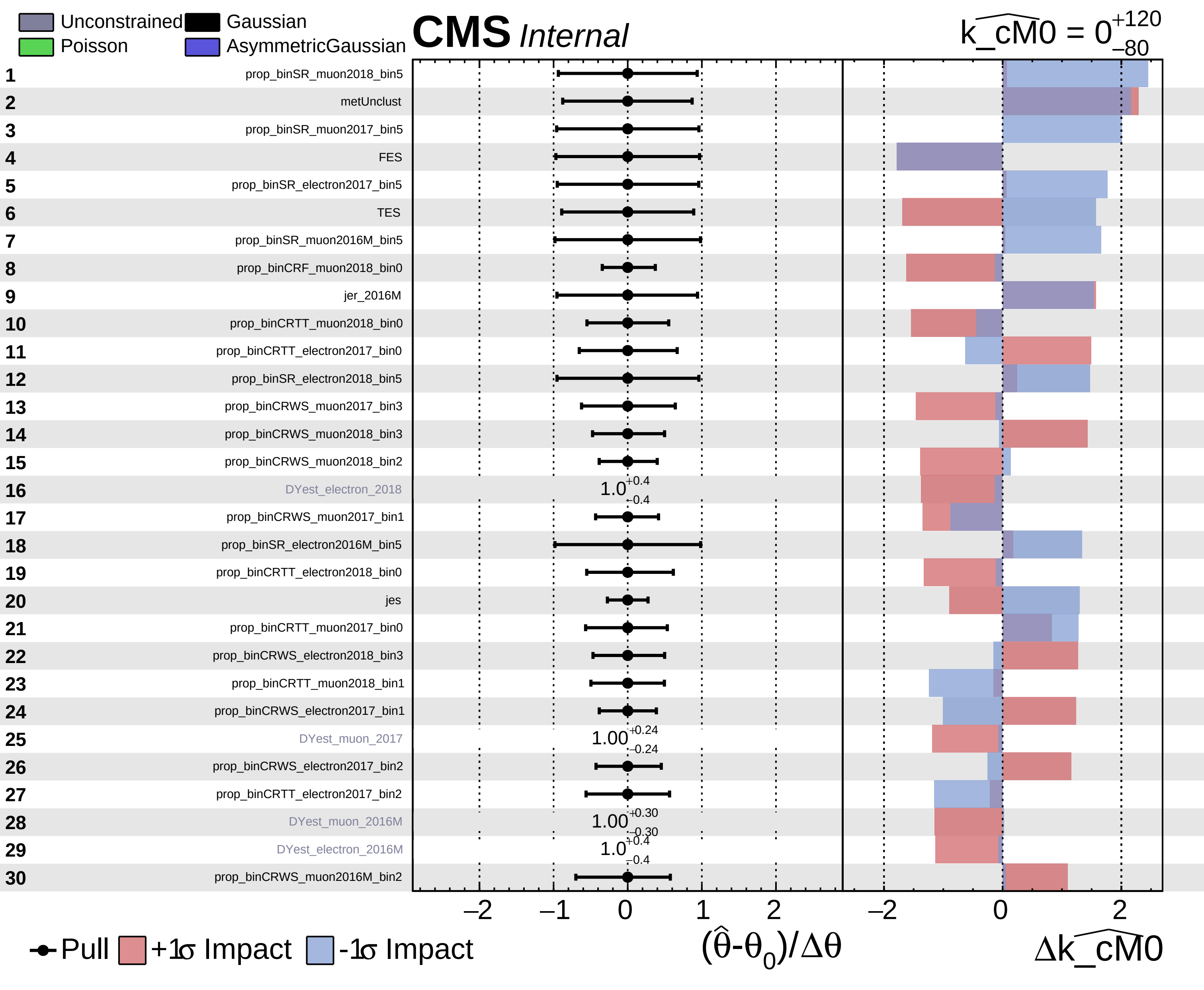
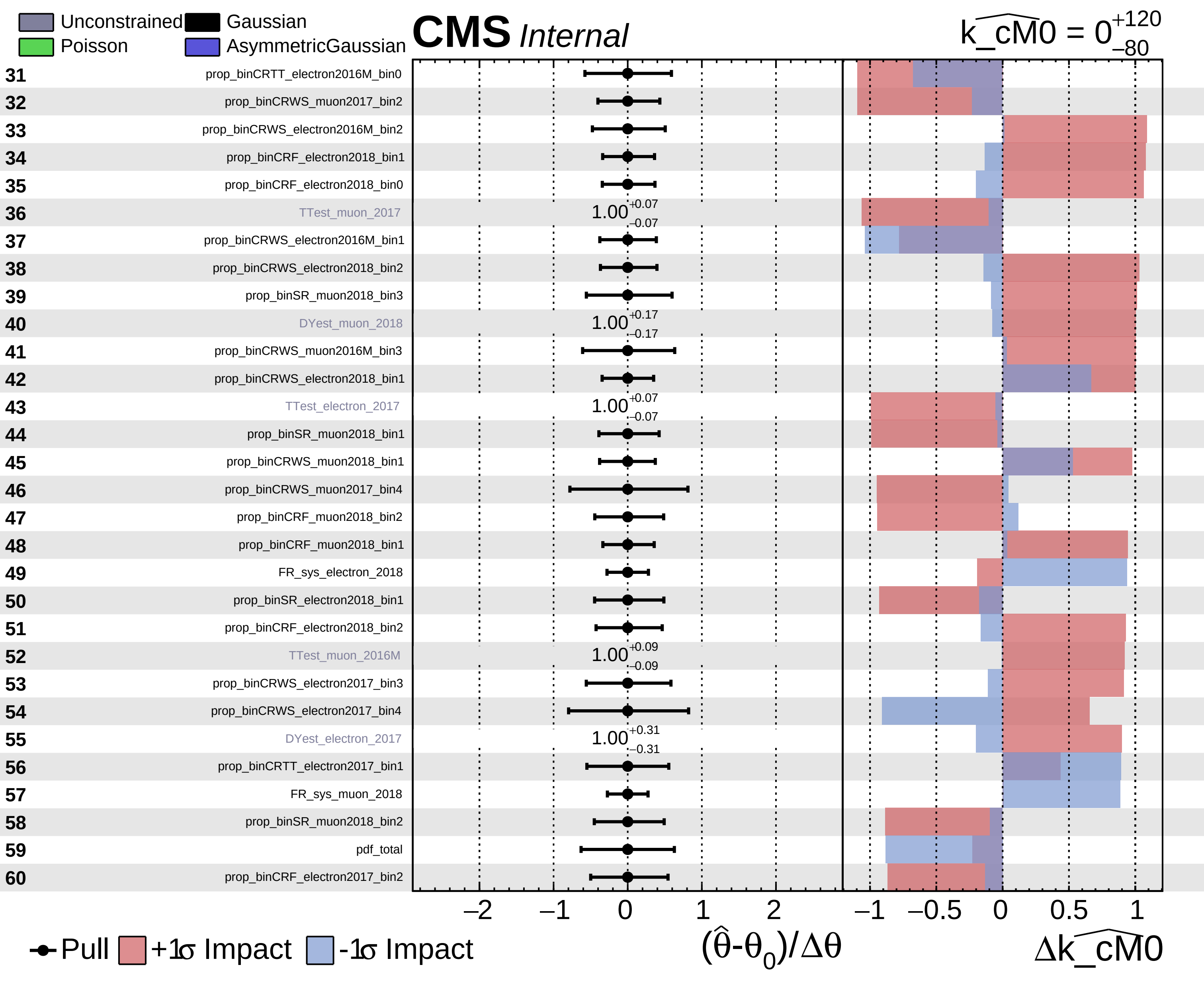
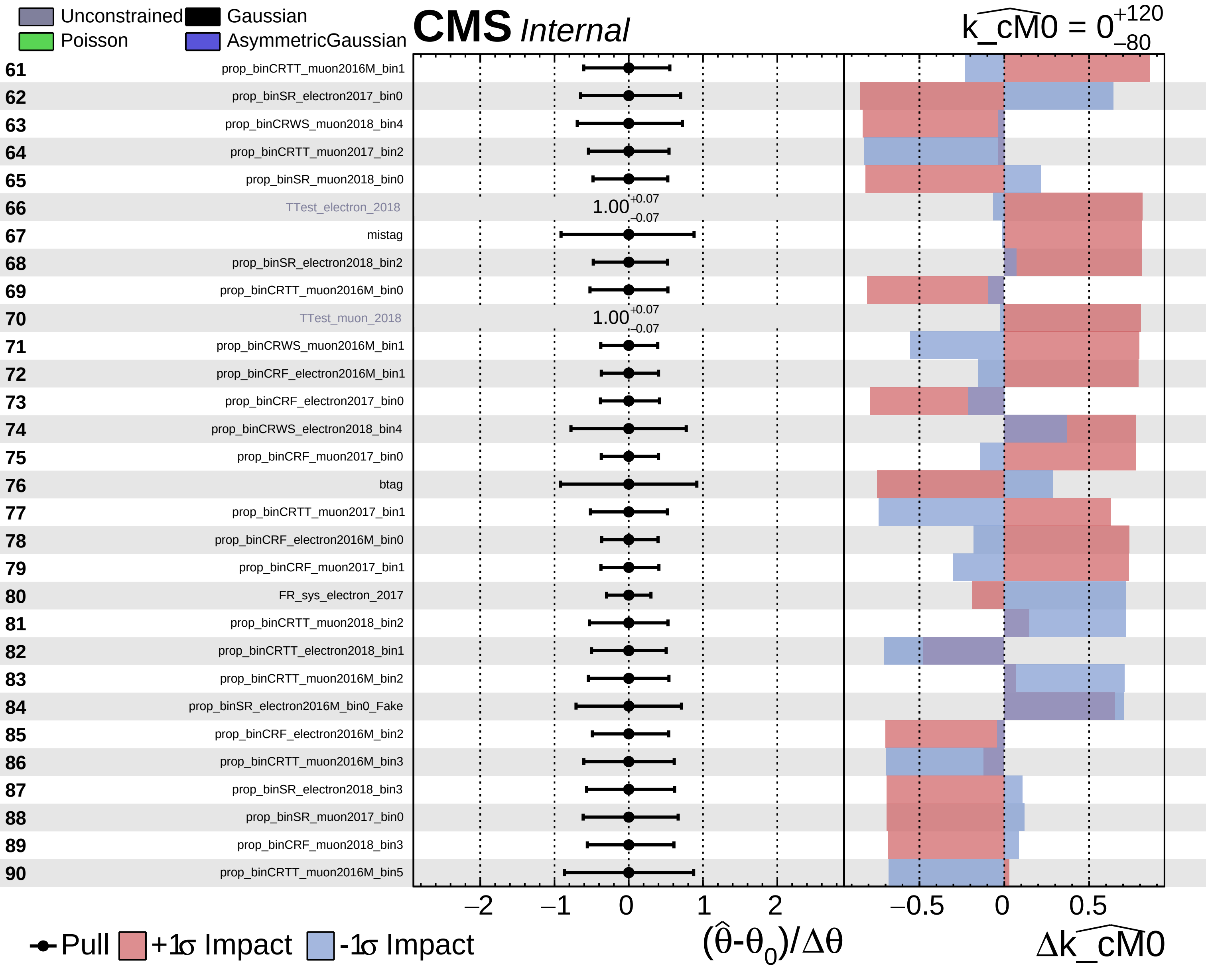
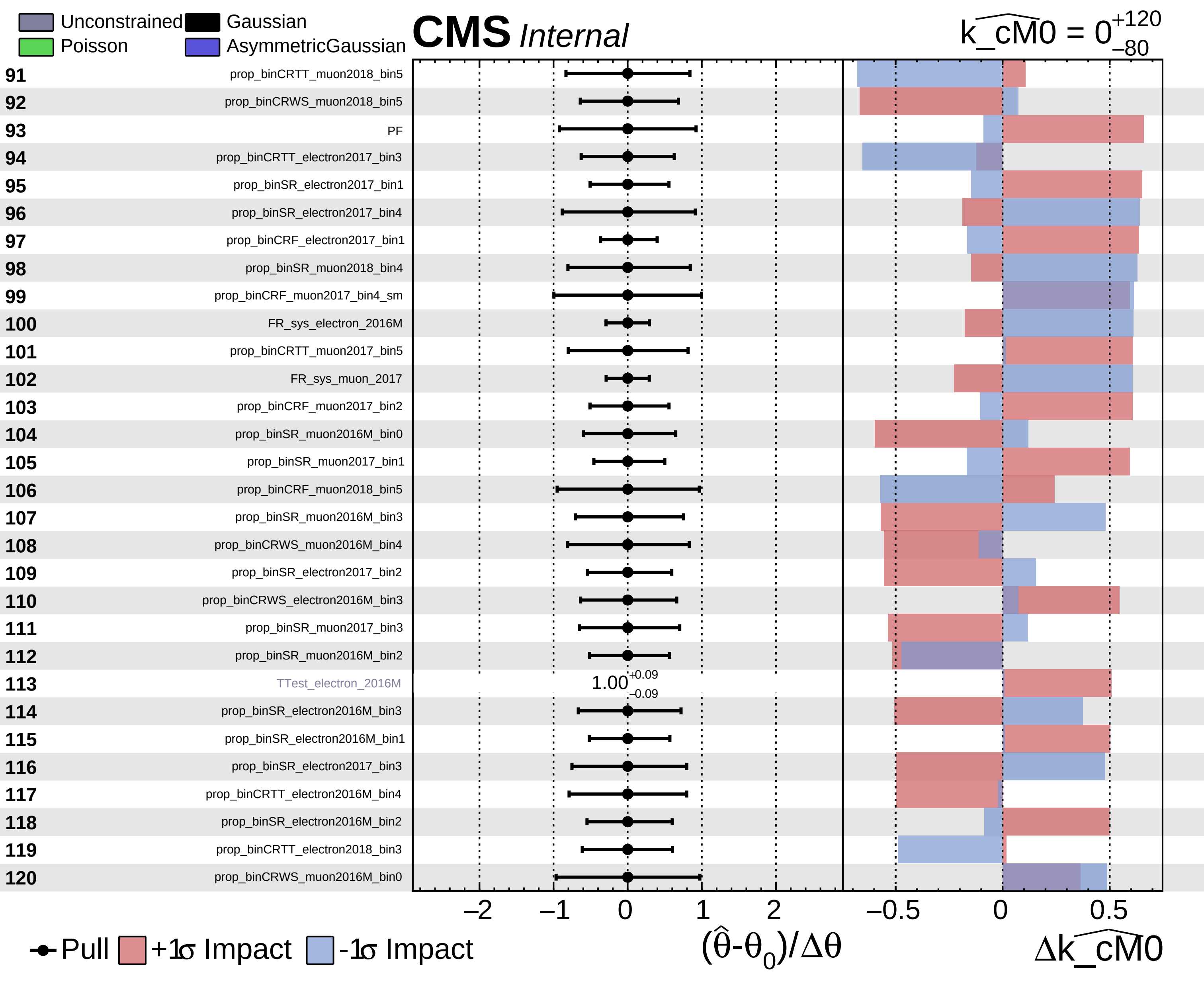
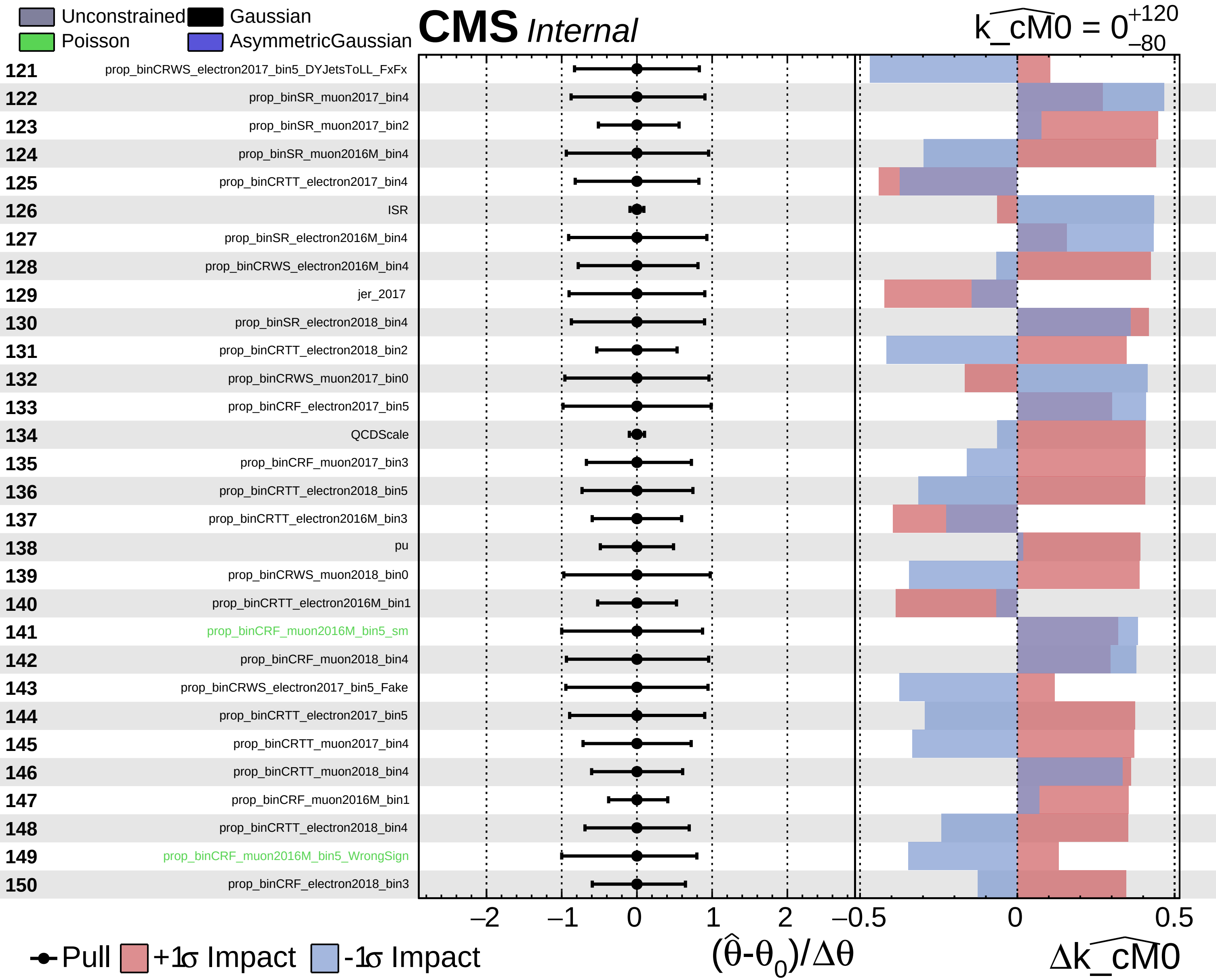

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0^{+120}_{-80}
1 prop_binSR_muon2018_bin5
2 metUnclust
3 prop_binSR_muon2017_bin5
4 FES
5 prop_binSR_electron2017_bin5
6 TES
7 prop_binSR_muon2016M_bin5
8 prop_binCRF_muon2018_bin0
9 jer_2016M
10 prop_binCRTT_muon2018_bin0
11 prop_binCRTT_electron2017_bin0
12 prop_binSR_electron2018_bin5
13 prop_binCRWS_muon2017_bin3
14 prop_binCRWS_muon2018_bin3
15 prop_binCRWS_muon2018_bin2
16 DYest_electron_2018
$1.0^{+0.4}_{-0.4}$
17 prop_binCRWS_muon2017_bin1
18 prop_binSR_electron2016M_bin5
19 prop_binCRTT_electron2018_bin0
20 jes
21 prop_binCRTT_muon2017_bin0
22 prop_binCRWS_electron2018_bin3
23 prop_binCRTT_muon2018_bin1
24 prop_binCRWS_electron2017_bin1
25 DYest_muon_2017
$1.00^{+0.24}_{-0.24}$
26 prop_binCRWS_electron2017_bin2
27 prop_binCRTT_electron2017_bin2
28 DYest_muon_2016M
$1.00^{+0.30}_{-0.30}$
29 DYest_electron_2016M
$1.0^{+0.4}_{-0.4}$
30 prop_binCRWS_muon2016M_bin2
−2 −1 0 1 2
−2 0 2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



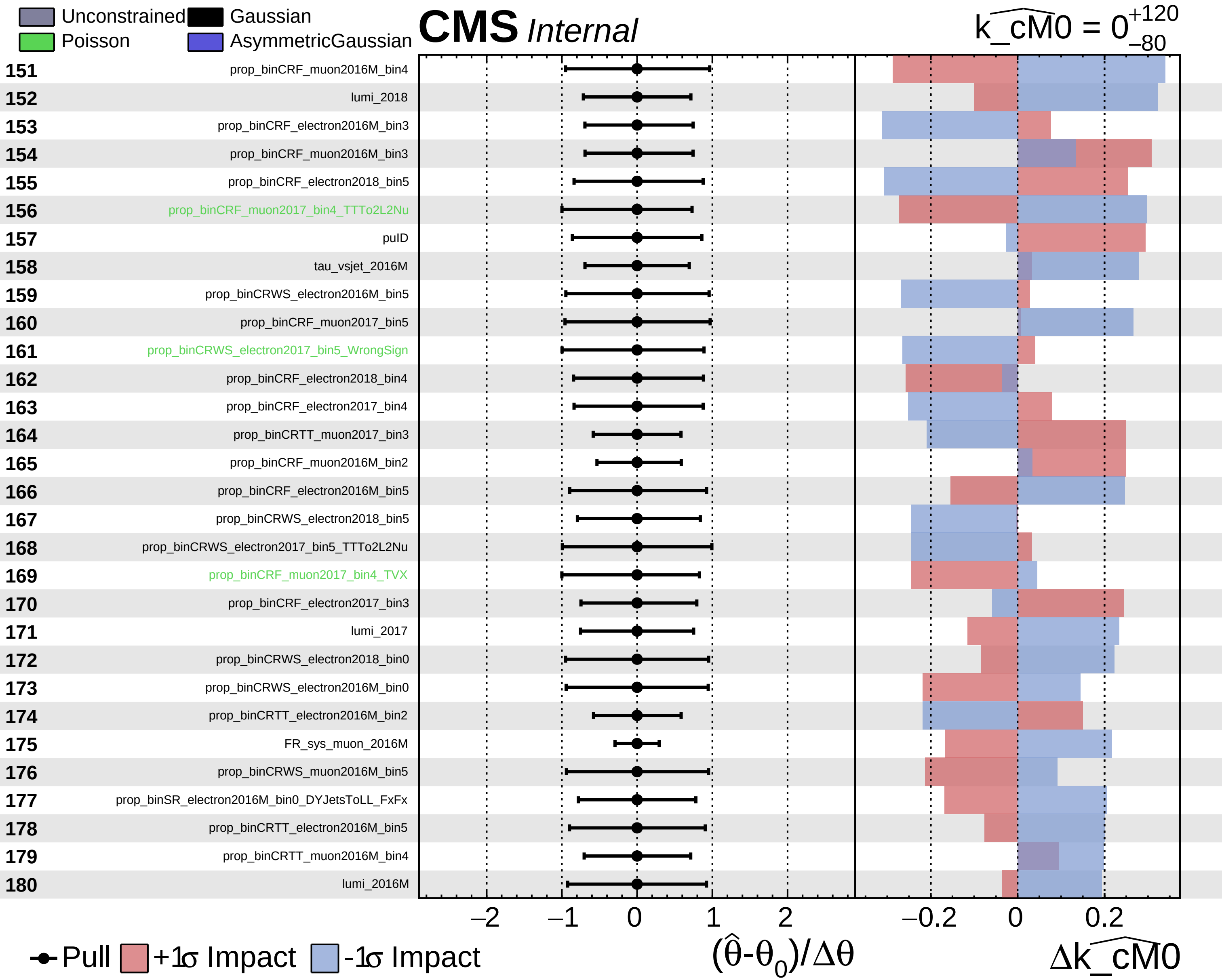
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
31 prop_binCRTT_electron2016M_bin0
32 prop_binCRWS_muon2017_bin2
33 prop_binCRWS_electron2016M_bin2
34 prop_binCRF_electron2018_bin1
35 prop_binCRF_electron2018_bin0
36 TTest_muon_2017 $1.00^{+0.07}_{-0.07}$
37 prop_binCRWS_electron2016M_bin1
38 prop_binCRWS_electron2018_bin2
39 prop_binSR_muon2018_bin3
40 DYest_muon_2018 $1.00^{+0.17}_{-0.17}$
41 prop_binCRWS_muon2016M_bin3
42 prop_binCRWS_electron2018_bin1
43 TTest_electron_2017 $1.00^{+0.07}_{-0.07}$
44 prop_binSR_muon2018_bin1
45 prop_binCRWS_muon2018_bin1
46 prop_binCRWS_muon2017_bin4
47 prop_binCRF_muon2018_bin2
48 prop_binCRF_muon2018_bin1
49 FR_sys_electron_2018
50 prop_binSR_electron2018_bin1
51 prop_binCRF_electron2018_bin2
52 TTest_muon_2016M $1.00^{+0.09}_{-0.09}$
53 prop_binCRWS_electron2017_bin3
54 prop_binCRWS_electron2017_bin4
55 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
56 prop_binCRTT_electron2017_bin1
57 FR_sys_muon_2018
58 prop_binSR_muon2018_bin2
59 pdf_total
60 prop_binCRF_electron2017_bin2
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +120 −80
61 prop_binCRTT_muon2016M_bin1
62 prop_binSR_electron2017_bin0
63 prop_binCRWS_muon2018_bin4
64 prop_binCRTT_muon2017_bin2
65 prop_binSR_muon2018_bin0
66 TTest_electron_2018 1.00 +0.07 −0.07
67 mistag
68 prop_binSR_electron2018_bin2
69 prop_binCRTT_muon2016M_bin0
70 TTest_muon_2018 1.00 +0.07 −0.07
71 prop_binCRWS_muon2016M_bin1
72 prop_binCRF_electron2016M_bin1
73 prop_binCRF_electron2017_bin0
74 prop_binCRWS_electron2018_bin4
75 prop_binCRF_muon2017_bin0
76 btag
77 prop_binCRTT_muon2017_bin1
78 prop_binCRF_electron2016M_bin0
79 prop_binCRF_muon2017_bin1
80 FR_sys_electron_2017
81 prop_binCRTT_muon2018_bin2
82 prop_binCRTT_electron2018_bin1
83 prop_binCRTT_muon2016M_bin2
84 prop_binSR_electron2016M_bin0_Fake
85 prop_binCRF_electron2016M_bin2
86 prop_binCRTT_muon2016M_bin3
87 prop_binSR_electron2018_bin3
88 prop_binSR_muon2017_bin0
89 prop_binCRF_muon2018_bin3
90 prop_binCRTT_muon2016M_bin5
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

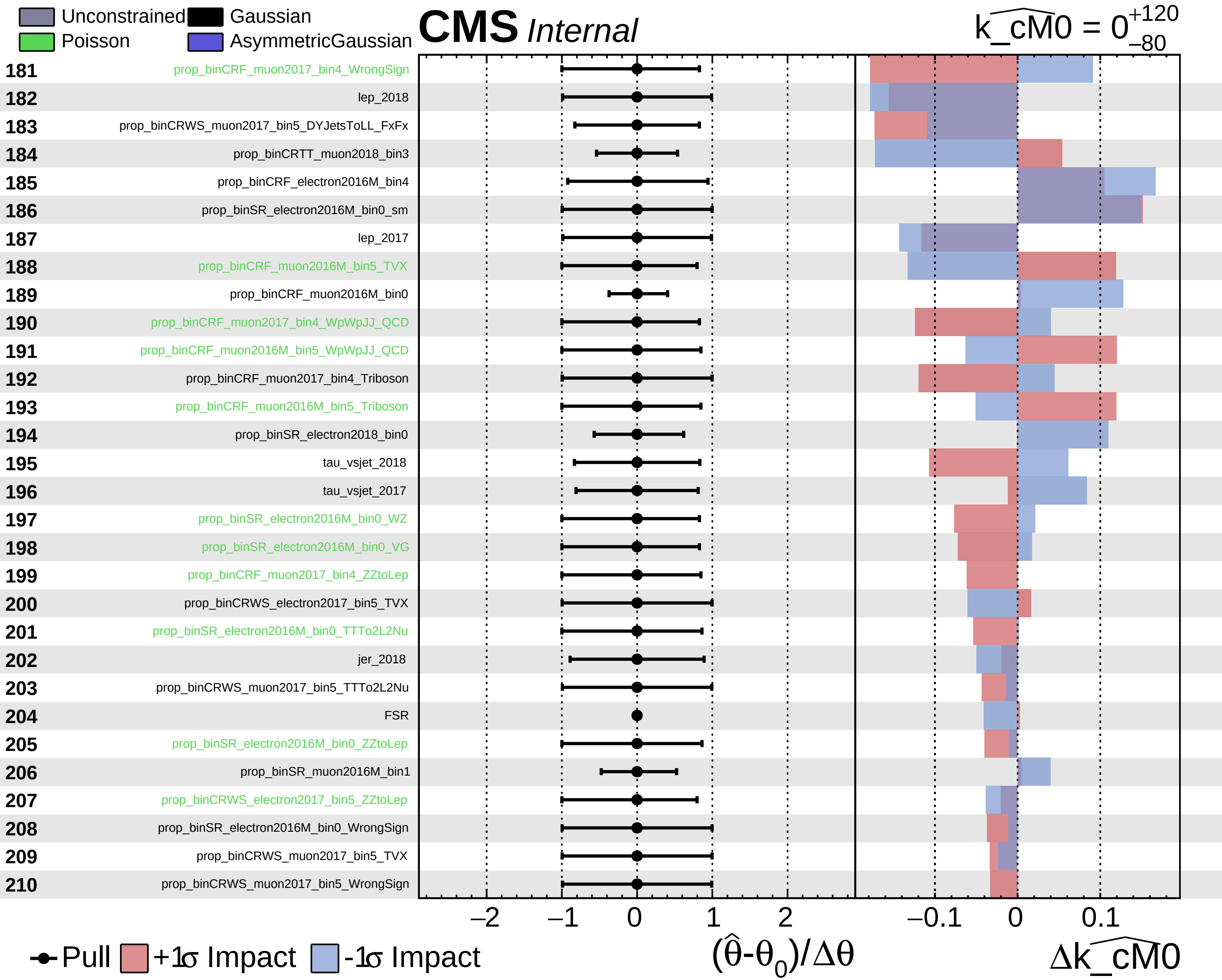

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
91 prop_binCRTT_muon2018_bin5
92 prop_binCRWS_muon2018_bin5
93 PF
94 prop_binCRTT_electron2017_bin3
95 prop_binSR_electron2017_bin1
96 prop_binSR_electron2017_bin4
97 prop_binCRF_electron2017_bin1
98 prop_binSR_muon2018_bin4
99 prop_binCRF_muon2017_bin4_sm
100 FR_sys_electron_2016M
101 prop_binCRTT_muon2017_bin5
102 FR_sys_muon_2017
103 prop_binCRF_muon2017_bin2
104 prop_binSR_muon2016M_bin0
105 prop_binSR_muon2017_bin1
106 prop_binCRF_muon2018_bin5
107 prop_binSR_muon2016M_bin3
108 prop_binCRWS_muon2016M_bin4
109 prop_binSR_electron2017_bin2
110 prop_binCRWS_electron2016M_bin3
111 prop_binSR_muon2017_bin3
112 prop_binSR_muon2016M_bin2
113 TTest_electron_2016M
$1.00^{+0.09}_{-0.09}$
114 prop_binSR_electron2016M_bin3
115 prop_binSR_electron2016M_bin1
116 prop_binSR_electron2017_bin3
117 prop_binCRTT_electron2016M_bin4
118 prop_binSR_electron2016M_bin2
119 prop_binCRTT_electron2018_bin3
120 prop_binCRWS_muon2016M_bin0
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

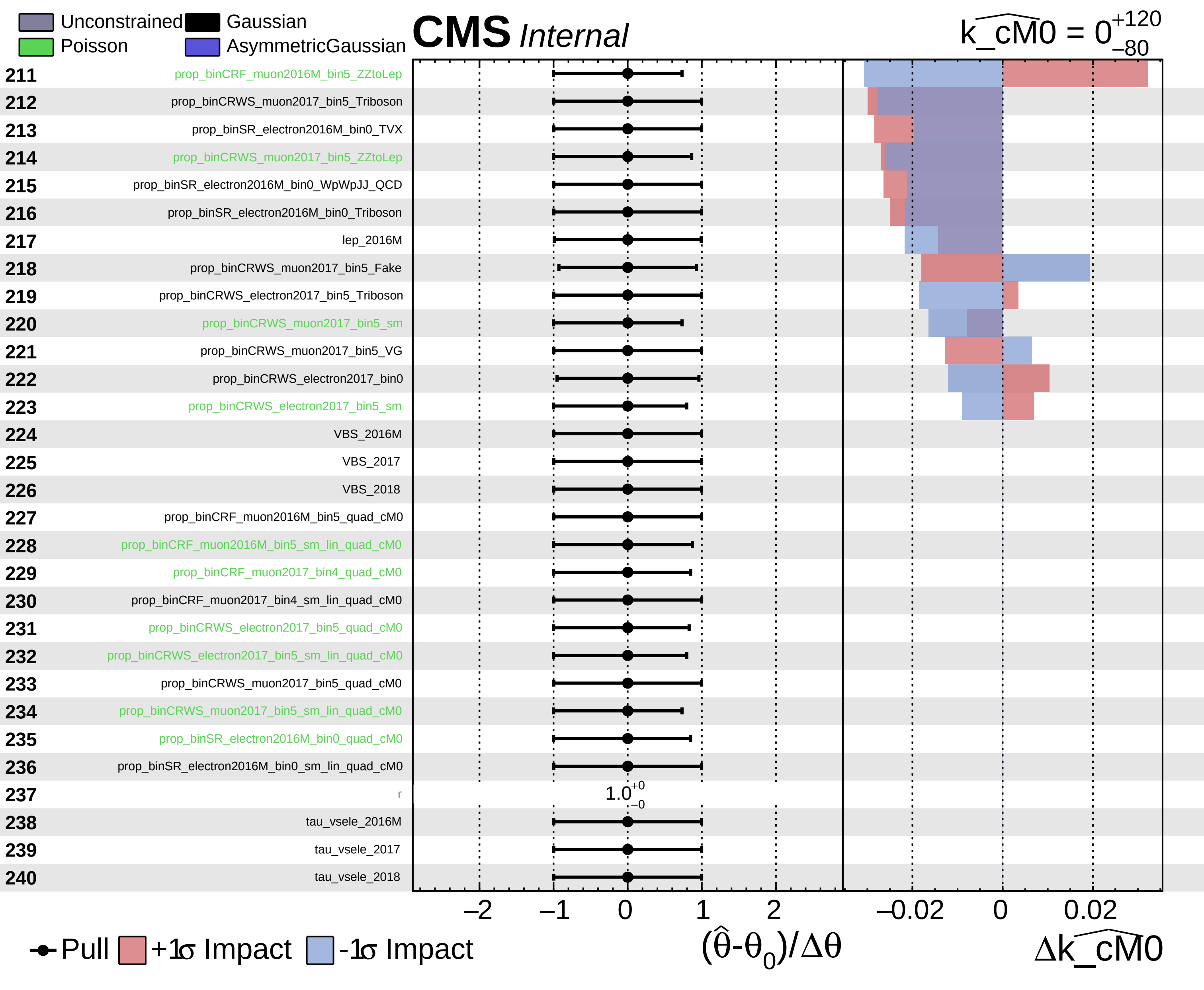


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
121 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
122 prop_binSR_muon2017_bin4
123 prop_binSR_muon2017_bin2
124 prop_binSR_muon2016M_bin4
125 prop_binCRTT_electron2017_bin4
126 ISR
127 prop_binSR_electron2016M_bin4
128 prop_binCRWS_electron2016M_bin4
129 jer_2017
130 prop_binSR_electron2018_bin4
131 prop_binCRTT_electron2018_bin2
132 prop_binCRWS_muon2017_bin0
133 prop_binCRF_electron2017_bin5
134 QCDScale
135 prop_binCRF_muon2017_bin3
136 prop_binCRTT_electron2018_bin5
137 prop_binCRTT_electron2016M_bin3
138 pu
139 prop_binCRWS_muon2018_bin0
140 prop_binCRTT_electron2016M_bin1
141 prop_binCRF_muon2016M_bin5_sm
142 prop_binCRF_muon2018_bin4
143 prop_binCRWS_electron2017_bin5_Fake
144 prop_binCRTT_electron2017_bin5
145 prop_binCRTT_muon2017_bin4
146 prop_binCRTT_muon2018_bin4
147 prop_binCRF_muon2016M_bin1
148 prop_binCRTT_electron2018_bin4
149 prop_binCRF_muon2016M_bin5_WrongSign
150 prop_binCRF_electron2018_bin3
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

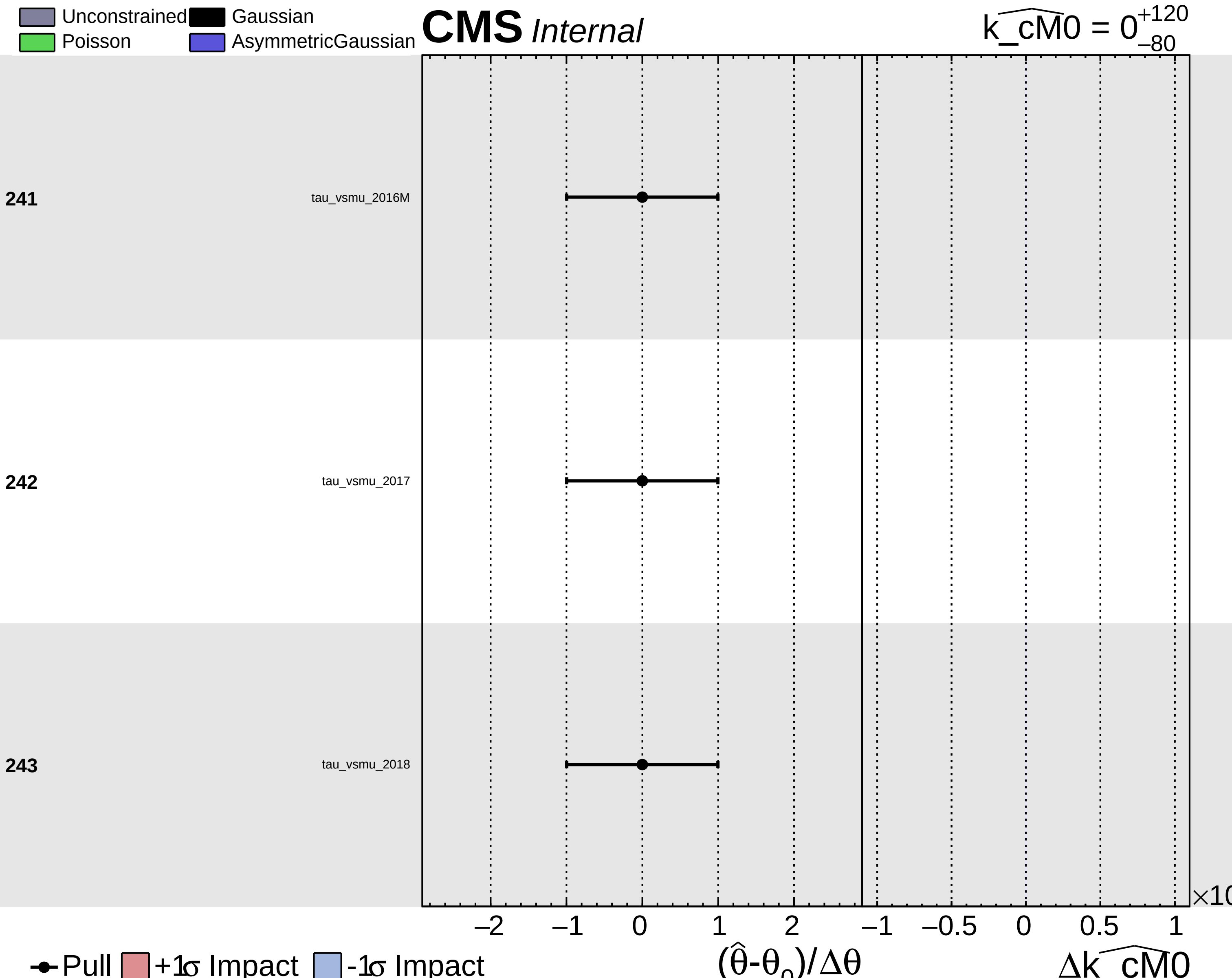


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
151 prop_binCRF_muon2016M_bin4
152 lumi_2018
153 prop_binCRF_electron2016M_bin3
154 prop_binCRF_muon2016M_bin3
155 prop_binCRF_electron2018_bin5
156 prop_binCRF_muon2017_bin4_TTTo2L2Nu
157 puID
158 tau_vsjet_2016M
159 prop_binCRWS_electron2016M_bin5
160 prop_binCRF_muon2017_bin5
161 prop_binCRWS_electron2017_bin5_WrongSign
162 prop_binCRF_electron2018_bin4
163 prop_binCRF_electron2017_bin4
164 prop_binCRTT_muon2017_bin3
165 prop_binCRF_muon2016M_bin2
166 prop_binCRF_electron2016M_bin5
167 prop_binCRWS_electron2018_bin5
168 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
169 prop_binCRF_muon2017_bin4_TVX
170 prop_binCRF_electron2017_bin3
171 lumi_2017
172 prop_binCRWS_electron2018_bin0
173 prop_binCRWS_electron2016M_bin0
174 prop_binCRTT_electron2016M_bin2
175 FR_sys_muon_2016M
176 prop_binCRWS_muon2016M_bin5
177 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
178 prop_binCRTT_electron2016M_bin5
179 prop_binCRTT_muon2016M_bin4
180 lumi_2016M
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
181 prop_binCRF_muon2017_bin4_WrongSign
182 lep_2018
183 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
184 prop_binCRTT_muon2018_bin3
185 prop_binCRF_electron2016M_bin4
186 prop_binSR_electron2016M_bin0_sm
187 lep_2017
188 prop_binCRF_muon2016M_bin5_TVX
189 prop_binCRF_muon2016M_bin0
190 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
191 prop_binCRF_muon2016M_bin5_WpWpJJ_QCD
192 prop_binCRF_muon2017_bin4_Triboson
193 prop_binCRF_muon2016M_bin5_Triboson
194 prop_binSR_electron2018_bin0
195 tau_vsjet_2018
196 tau_vsjet_2017
197 prop_binSR_electron2016M_bin0_WZ
198 prop_binSR_electron2016M_bin0_VG
199 prop_binCRF_muon2017_bin4_ZZtoLep
200 prop_binCRWS_electron2017_bin5_TVX
201 prop_binSR_electron2016M_bin0_TTTo2L2Nu
202 jer_2018
203 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
204 FSR
205 prop_binSR_electron2016M_bin0_ZZtoLep
206 prop_binSR_muon2016M_bin1
207 prop_binCRWS_electron2017_bin5_ZZtoLep
208 prop_binSR_electron2016M_bin0_WrongSign
209 prop_binCRWS_muon2017_bin5_TVX
210 prop_binCRWS_muon2017_bin5_WrongSign
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
211 prop_binCRF_muon2016M_bin5_ZZtoLep
212 prop_binCRWS_muon2017_bin5_Triboson
213 prop_binSR_electron2016M_bin0_TVX
214 prop_binCRWS_muon2017_bin5_ZZtoLep
215 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
216 prop_binSR_electron2016M_bin0_Triboson
217 lep_2016M
218 prop_binCRWS_muon2017_bin5_Fake
219 prop_binCRWS_electron2017_bin5_Triboson
220 prop_binCRWS_muon2017_bin5_sm
221 prop_binCRWS_muon2017_bin5_VG
222 prop_binCRWS_electron2017_bin0
223 prop_binCRWS_electron2017_bin5_sm
224 VBS_2016M
225 VBS_2017
226 VBS_2018
227 prop_binCRF_muon2016M_bin5_quad_cM0
228 prop_binCRF_muon2016M_bin5_sm_lin_quad_cM0
229 prop_binCRF_muon2017_bin4_quad_cM0
230 prop_binCRF_muon2017_bin4_sm_lin_quad_cM0
231 prop_binCRWS_electron2017_bin5_quad_cM0
232 prop_binCRWS_electron2017_bin5_sm_lin_quad_cM0
233 prop_binCRWS_muon2017_bin5_quad_cM0
234 prop_binCRWS_muon2017_bin5_sm_lin_quad_cM0
235 prop_binSR_electron2016M_bin0_quad_cM0
236 prop_binSR_electron2016M_bin0_sm_lin_quad_cM0
237 r
1.0^{+0}_{-0}
238 tau_vsele_2016M
239 tau_vsele_2017
240 tau_vsele_2018
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+120}_{-80}$
241
tau_vsmu_2016M
242
tau_vsmu_2017
243
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$